

Efficient Multimedia Computing: Unleashing the Power of AutoML

Debanjan Datta
Amazon Web Services

Gerald Friedland
Amazon Web Services, UC Berkeley

ABSTRACT

As the field of multimedia computing has grown rapidly, so has the need for larger datasets [5] and increased modeling capacity. Navigating this complex landscape often necessitates the use of sophisticated tools and cloud architectures, which all need to be addressed before the actual research commences. Recently, AutoML, an innovation previously exclusive to tabular data, has expanded to encompass multimedia data. This development has the potential to greatly streamline the research process, allowing researchers to shift their focus from model construction to the core content of their problems. In doing so, AutoML not only optimizes resource utilization but also boosts the reproducibility of results. The aim of this tutorial is to acquaint the multimedia community with AutoML technologies, underscoring their advantages and their practical applications in the field.

ACM Reference Format:

Debanjan Datta and Gerald Friedland. 2023. Efficient Multimedia Computing: Unleashing the Power of AutoML. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3581783.3613858>

1 TUTORIAL DESCRIPTION

Much of the time of a multimedia researcher is spent training and tuning models in a trial-and-error fashion, burning resources that could otherwise be used for more interesting tasks. This tutorial presents techniques to reduce this overhead by introducing AutoML for multimedia computing. It will start with a fundamental introduction to AutoML, its concepts, and benefits. This will be followed by a deep dive into the practical application of AutoML technologies for multimedia data, including image, audio, video, and AR/VR data. The tutorial will also feature hands-on examples using AWS SageMaker tools demonstrating how AutoML can streamline research processes, enhance model accuracy, and reduce costs associated with traditional machine learning models. The tutorial has two main parts which are summarized as follows.

2 THEORETICAL PART

The theoretical part presents the foundation of how AutoML works, what problems it solves and what its limits are.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0108-5/23/10.

<https://doi.org/10.1145/3581783.3613858>

AutoML is defined as a set of methods and processes that are designed to make Machine Learning accessible in a useful but efficient manner for non-Machine Learning experts so that ML can be leveraged for their tasks [3].

There are multiple steps to the process of AutoML, all or some of which are executed based on the problem and system design: (1) Data ingestion (2) Data cleaning or preparation (3) Feature engineering or extraction (4) Algorithm selection (5) Model training (6) Hyperparameter optimization

Each of the steps has its specific challenges. Feature extraction can be challenging, and complex but automated strategies that extract features for candidate set of models builds in efficiency. The algorithm selection is another step that requires practitioners to understand the data. Instead AutoML abstracts this through approaches such as meta-learning, and provides the user with a model with preset hyperparameter settings that are either tailored for that dataset or are known to perform well across a wide gamut of use cases. Hyperparameter optimization is another aspect that induces complexity, and autoML can effectively manage that.

AutoML abstracts these challenges and presents a trained and tuned model to the user that work best for the data. AutoML is able to approach machine learning problems in a different way than humans would. A prominent example for this is mentioned in [1] and reproduced in Figure 1. This simple, academic example shows how AutoML can approach solving XOR with a single neuron and succeed at it. AutoML is well suited to research problems faced by multimedia researchers, in harnessing the power of recent Deep Learning models. It takes out the element of figuring out which models may not fit the bill. Hosted AutoML at scale provides reliable, robust and scalable solutions. Rich multimedia data such as images, text, audio and video have been focus of deep learning research over the last decade, and AutoML allows multimedia researchers to leverage this progress. It ultimately allows the multimedia researcher to focus on the application and on the specific problem, rather than spending much of the time developing machine learning algorithms in a trial-and-error fashion.

3 PRACTICAL PART

3.1 Introduction to AWS Sagemaker

This section introduces the users to AWS Sagemaker and the associated services. AWS Sagemaker is a cloud-based service that provides AutoML as a service, that allows users to experiment, explore and deploy ML models to solve problems.

3.2 Understanding How AutoML makes the Machine Learning process more efficient

This section provides an overview of the open source tools that are available for solving multimedia tasks. Specifically we consider 2 use cases, namely image and text classification.

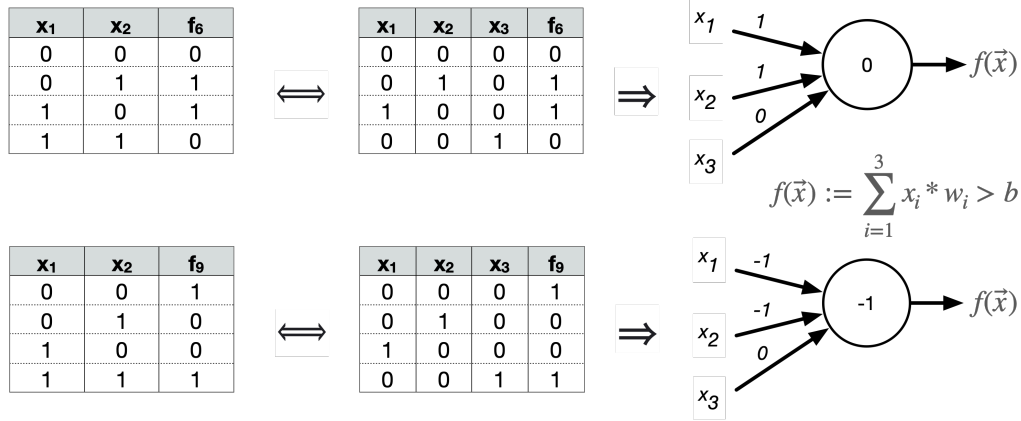


Figure 1: As prominently criticized in [4], the 2-variable Boolean functions f_6 (XOR) and f_9 (equality) cannot be modeled by a single 2 – input neuron. However, AutoML technologies can approach problems differently and, for example, re-code the input data (here: into one-hot encoding). By doing that, the single neuron’s capacity can be increased by another input weight and so it is able to model both functions. Image 16.2 from [1].

We show two examples, of how a pipeline for performing these tasks is set up using open source tools that are commonly used by researchers in machine learning. This entails understanding (i) what libraries are compatible, (ii) how to set up the execution environment, (iii) setting up the data pipeline, including how to pre-process the data and format it for the model, (iv) and training the model including data management for batching to adhere to memory constraints and choice of optimization strategy. While there are knowledge sources openly and readily available, they require a significant time investment and a rather steep learning curve. The question thus arises, could these complexities be abstracted? Abstraction can alleviate the aforementioned issues, yet providing multimedia researchers a way to achieve their goals using state-of-the-art machine learning models. AutoML provides this abstraction, and manages performance vs. cost trade offs effectively.

AutoML for Image Classification: We present an example on how AWS’s Sagemaker and associated services can perform image classification. The widely used, and highly relevant publicly available CIFAR100 dataset is chosen for this purpose.

AutoML for Text Classification: Sentiment classification is a use-case of text classification. We provide a walk-through of how this can be performed using Sagemaker AutoML, on a publicly available dataset viz. AWS Reviews dataset.

4 CONCLUSION

The prowess of AutoML lies in its capacity to transform an often tedious, resource-consuming process into an automated, optimized journey. By entrusting intricate tasks like feature extraction, algorithm selection, and hyperparameter optimization to AutoML, researchers can realign their focus to the core issues of their domain. The tutorial’s hands-on exercises with AWS Sagemaker aptly showcase how the synthesis of AutoML and multimedia can enhance model accuracy, optimize resource allocation, and drastically reduce costs. Furthermore, the applicability of AutoML in multimedia computing, spanning from image to text classification, underscores

its versatility. AutoML bridges the chasm between raw data and actionable insights, without burdening researchers with the intricacies of model selection and tuning. We believe the proliferation of AutoML in the realm of multimedia computing marks a seminal moment for the field. It is not just a tool but a paradigm shift, championing efficiency, reproducibility, and robustness.

SHORT BIOS OF THE ORGANIZERS

Dr. Debanjan Datta is an Applied Scientist at Amazon AWS, with a wealth of experience in AutoML and its applications, and is a part AWS Sagemaker Low Code No Code organization. His research interests include data mining and ML, with a focus on anomaly detection, explainability, tabular and multimedia data. He received his PhD from Virginia Tech in 2022, where his research focus was anomaly detection and explainability in large-scale tabular data.

Dr. Gerald Friedland is adjunct faculty at UC Berkeley and the Principal Scientist for AutoML at Amazon AWS. He has published extensively (see Google Scholar) in the field of multimedia signal processing, multimedia content analysis, and machine learning. His track record includes winning the ACM Multimedia Grand Challenge, being the author of the 2012 textbook “Introduction to Multimedia Computing” with Ramesh Jain [2] and being the co-creator of the YFCC100M corpus [5]. His 2023 textbook [1] is to be published by Springer-Nature by the conference date. His PhD is from Freie Universitaet Berlin, 2006.

REFERENCES

- [1] Gerald Friedland. *Information-based Machine Learning: Data Science as an Engineering Discipline*. Springer-Nature, 2023.
- [2] Gerald Friedland and Ramesh Jain. *Multimedia computing*. Cambridge University Press, 2014.
- [3] Xin He, Kaiyong Zhao, and Xiaowen Chu. Auttml: A survey of the state-of-the-art. *Knowledge-Based Systems*, 212:106622, 2021.
- [4] M. Minsky and S. Papert. *Perceptrons: An Introduction to Computational Geometry*. MIT Press, 1969.
- [5] Bart Thomee, David A Shamma, Gerald Friedland, Benjamin Elizalde, Karl Ni, Douglas Poland, Damian Borth, and Li-Jia Li. Yfcc100m: The new data in multimedia research. In *Proceedings of the 2016 ACM on Conference on Multimedia Conference*, pages 242–251. ACM, 2016.